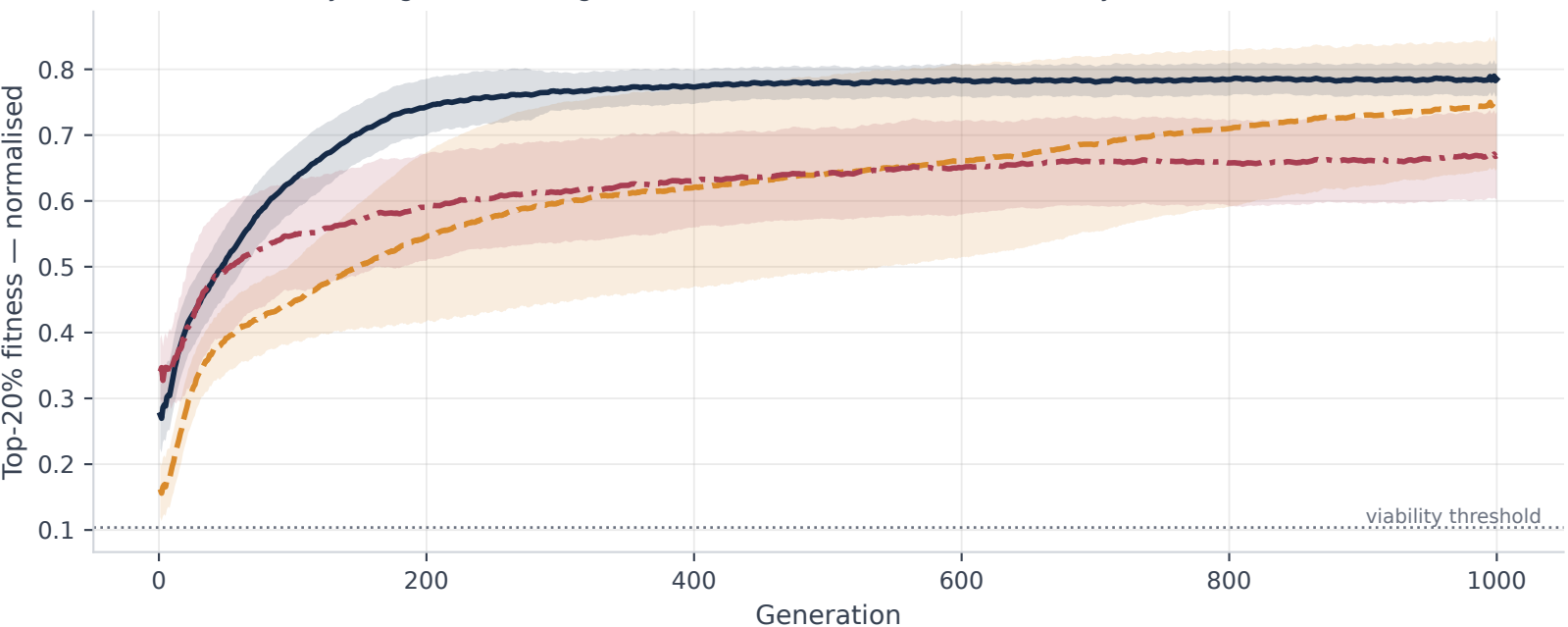


Top-20% fitness — Predator environment (60 roaming, drain 5) · 300 runs across 3 conditions

Mean over runs · shaded ± 1 std across runs · smoothed 15 gens · converged = last 50 gens · y cropped to [0.07, 0.89]
rescaled by the global raw range [2.00, 24.92] — one transform for every condition and environment



converged mean ± 1 std across runs

Evolution (EA only)
0.7421 $\pm$ 0.0961 (n = 100)

Learning (EA + in-life RL)
0.7849 $\pm$ 0.00866 (n = 100)

Pure RL (no EA)
0.6676 $\pm$ 0.0568 (n = 100)